DEPT. OF ELECTRONIC & TELECOMMUNICATION ENGINEERING · UNIVERSITY OF MORATUWA

# Closing the Deictic Loop

Bidirectional deictic grounding for gesture-conditioned vision-language-action navigation on a quadruped

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Project portfolio

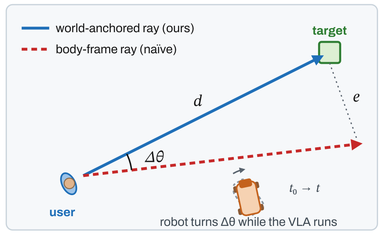
1 · MOTIVATION

## Pointing is brief. VLA inference is slow.

**Pointing is brief:** about 1–2 s, and tied to the pose the robot had when it saw it.

**VLA inference is slow.** If the robot is still turning and keeps the ray in its body frame, the ray ends up wrong by exactly the rotation $\Delta\theta$ made during inference — a geometric identity, not noise.

**Research gap.** No existing system combines pointing input, VLA-driven quadruped locomotion and a robot→human grounding channel.




Un-anchored error ≡ $|\Delta\theta|$; miss $e \approx d \cdot \tan\Delta\theta$. LEGS-POMDP: 14.4° ≈ 1.3 m off at 5 m.

2 · CONTRIBUTIONS

## What we add

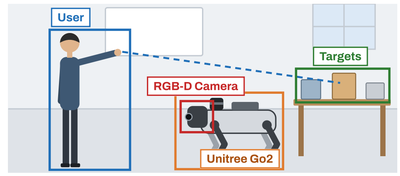
1. **Geometric identity:** an un-anchored pointing ray is off by exactly $|\Delta\theta|$, the rotation during inference.
2. **Gesture memory buffer:** rays anchored in the world frame (TF2) at capture, re-projected at use.
3. **Bidirectional grounding:** the robot turns, asks, and takes a confirm or a correction.
4. **Gesture-grounding metrics** with a 1–5 s latency sweep.

3 · DATA COLLECTION

## Lab and simulation setups




(a) Captured frame: user pointing (640×480, HFOV 69.5°)




(b) Lab setup — Unitree Go2 with RGB-D camera, Velodyne LiDAR and Jetson AGX Orin; user points from distance $d$



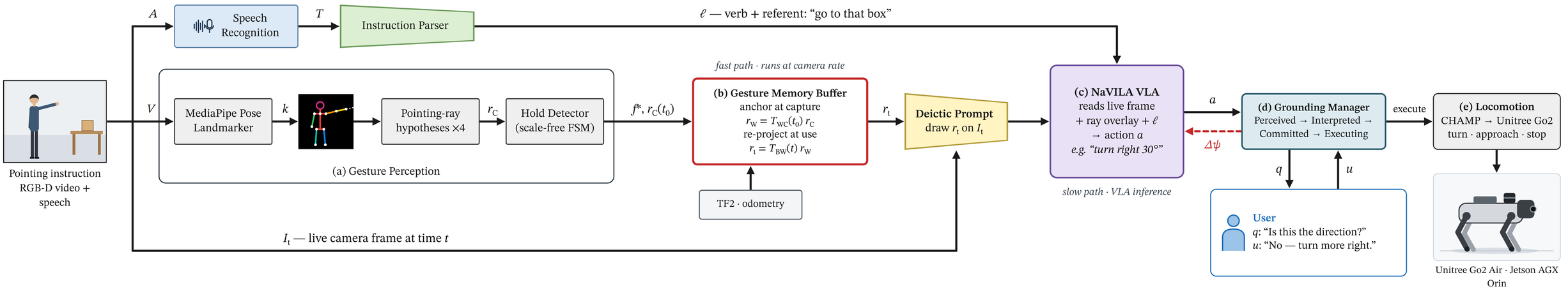

(c) Simulation — Gazebo + CHAMP + HuNavSim pedestrians

**Sweeps:** distance · target layout · injected latency $\tau$ = 1–5 s

**(d) Logged per trial** RGB-D frames $V$ · Speech audio $A$ · Pose keypoints $k$ · TF2 / odometry · Velodyne LiDAR · GT target pose

4 · PROPOSED METHOD · SYSTEM ARCHITECTURE

## A fast gesture path feeding a slow VLA

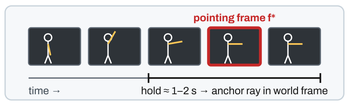



5 · FROM POINTING TO APPROACH

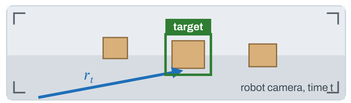
## From a point to a stop in six steps




**1 Point & speak**
User points at a box and says "go to that box".



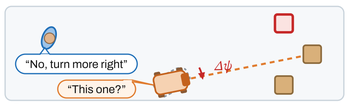

**2 Perceived**
Hold detected; pointing frame $f^*$ kept, ray anchored in the world frame.




**3 Interpreted**
NaVILA reads the live frame with $r_t$ drawn in and picks the box.




**4 Committed**
Robot turns to face its pick and asks "Is this the direction?"



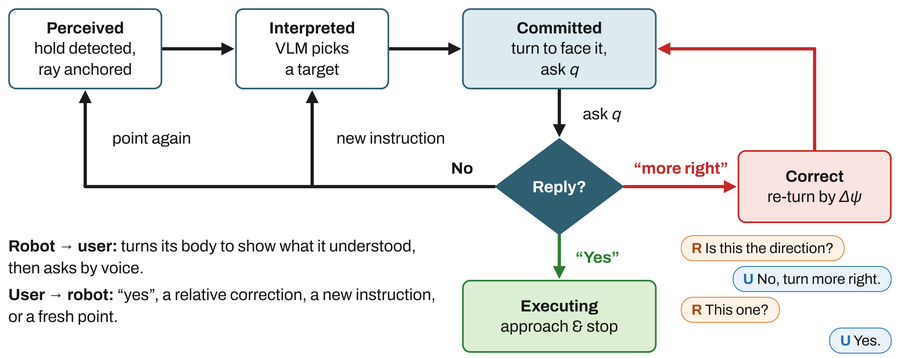

**5 Corrected**
"No, turn more right." Robot adds $\Delta\psi$, re-turns and asks again.




**6 Executing**
"Yes." Robot walks to the box and stops in front of it.

6 · BIDIRECTIONAL GROUNDING LOOP

## The robot shows what it understood, then asks




**Robot → user:** turns its body to show what it understood, then asks by voice.

**User → robot:** "yes", a relative correction, a new instruction, or a fresh point.

7 · EVALUATION

## Four metrics

**Deictic heading error**
final heading vs true target bearing (°)

**Instruction capture rate**
pointing events caught (%)

**Target selection accuracy**
commits to the right object (%)

**Latency-conditioned degradation**
all metrics vs injected $\tau$ = 1–5 s

**Compare:** body-frame ray (naïve) vs world-anchored ray (ours).

**Where:** HuNavSim in Gazebo, then the real Go2.

**PLATFORM** Unitree Go2 Air · Jetson AGX Orin 32 GB · Velodyne LiDAR · RGB-D camera · ROS 2 Humble · TF2 · MediaPipe · NaVILA · CHAMP · Gazebo · HuNavSim

**RELATED WORK** NaVILA · GIVE · LEGS-POMDP · GesVLA · TIC-VLA · HuNavSim